Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2025

Course: B.Tech | *Time: 03 Hours*

Code: 104503 | *Subject: Linear Control System* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.

(ii) There are NINE questions in this paper.

(iii) Attempt FIVE questions in all.

(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven questions only)*** **[2 x 7 = 14]**

(a) At high frequency, the asymptotic slope of the Bode magnitude plot for the transfer function $G(s) = \frac{100}{s(1+0.2s)(1+0.02s)}$

(i) -30 dB/decade (iii) -60 dB/decade

(ii) $+60$ dB/decade (iv) $+20\ dB/decade$

(b) The characteristic equation of a system is $s^3 + 5s^2 + 8s + K = 0$ for marginal stability, the value of K is

(i) 8 (iii) 40

(ii) 20 (iv) 50

(c) For the transfer function $G(s) = \frac{1}{(s+1)}$ the magnitude at $\omega = 1$ rad/s is

(i) $0\ dB$ (iii) $-6\ dB$

(ii) $-3\ dB$ (iv) $-10\ dB$

(d) For a system with an open-loop transfer function $G(s)H(s) = \frac{K}{s(s+2)(s+4)}$ the number of asymptotes in the root locus is

(i) 1 (iii) 3

(ii) 2 (iv) 4

(e) The root locus technique graphically shows the movement of

(i) Zeros only (iii) Poles only

(ii) Closed-loop poles with varying gain (iv) Open-loop zeros with varying damping

(f) A system is completely controllable if

(i) The rank of the controllability matrix is less than the system order (iii) The rank of the controllability matrix equals the system order

(ii) Determinant of the system matrix is zero (iv) All eigenvalues are positive

(g) For a unity feedback system with an open-loop transfer function $G(s) = \frac{20}{s(s+4)}$ the velocity error constant, K_v

(i) 2 (iii) 5

(ii) 4 (iv) 20

(h) For a minimum phase system, all zeros lie in

(i) Right half plane (iii) Imaginary axis only

(ii) Left half plane (iv) Origin only

(i) Which of the following devices converts shaft angular position into an electrical signal used for error detection in servo systems?

(i) Tacho generator (iii) Stepper motor

(ii) Synchro (iv) Magnetic amplifier

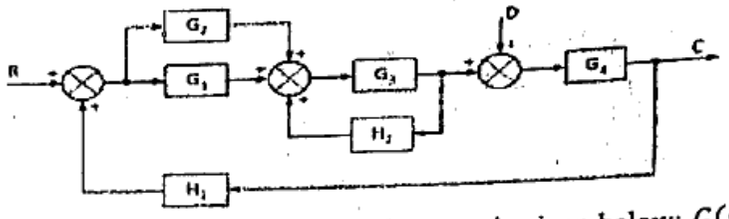
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(j) The steady-state error for a Type-1 system subjected to unit step input is
(i) Infinite (iii) Zero
(ii) Unity (iv) Depends on damping ratio

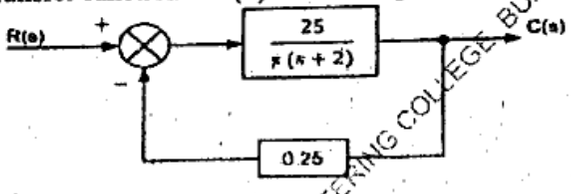
Q.2 The open-loop transfer function of a control system is given below: $G(s)H(s) = \frac{30}{s(1+0.5s)(1+0.08s)}$. Draw the Bode plot and determine: P.M., G.M., and comment on the closed-loop stability of the system. [14]

Q.3 (a) Find C/R, C/D, and the total output for the system whose block diagram is shown below [7]



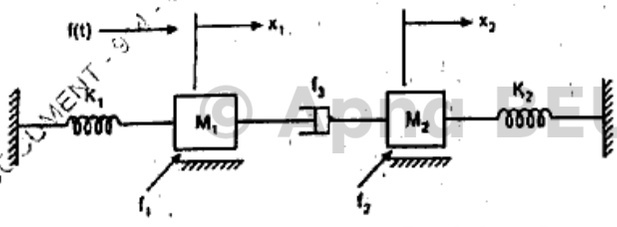
(b) The open-loop transfer function of a control system is given below: $G(s)H(s) = \frac{K}{s(s+6)(s^2+4s+13)}$. Sketch the root locus and find the range of K for the system to be stable. [7]

Q.4 (a) Determine the sensitivity of the overall transfer function at $\omega = 1\ rad/sec$ w.r.t. [7]
(i) Forward path transfer function (ii) Feedback path transfer function



(b) Obtain the time response of the system given below: $\dot{x} = Ax$, Where $A = \begin{bmatrix} 0 & 1 \\ -2 & 0 \end{bmatrix}$; $x(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $y = [1 \quad -1]\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. [7]

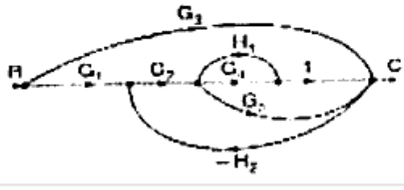
Q.5 (a) For the given translational mechanical system, find the Force–Voltage (F–V) analogy. [7]



(b) Check the controllability of the system defined by the state matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix}$, $B = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $C = [10 \quad 5 \quad 1]$. [7]

Q.6 The open-loop transfer function of a control system is given below: $G(s) = \frac{K}{s(1+0.2s)}$. Design a suitable compensator such that the system will have $K_v = 10$ and P.M. $= 50°$. [14]

Q.7 (a) Find the transfer function C/R [7]



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(b) Using Nyquist stability criteria, examine the closed-loop stability of a control system whose open-loop transfer function is given below: $G(s)H(s) = \frac{K}{s(1+sT)}$. [7]

Q.8 (a) The open-loop transfer function of a control system is given by $G(s) = \frac{K}{s(1+sT_1)(1+sT_2)}$. Apply Routh-Hurwitz criteria to determine the values of K in terms of T_1 & T_2 for the system to be stable. [7]

(b) A second-order system is represented by a transfer function given below $\frac{\theta_o(s)}{T(s)} = \frac{1}{Js^2+fs+k}$, Where θ_o the proportional output and T is the input torque. A step input of 10 Nm is applied to the system, and the test results are given below: $M_p = 6\ \%, t_p = 1\text{sec}$, and steady state output is 0.5 rad. Determine the values of J, f & k. [7]

[7 x 2 = 14]

Q.9 Write short notes on ***any two*** of the following:

(a) Absolute, conditional, and marginal stability with suitable examples

(b) Nyquist stability criterion

(c) P, PI, and PID controllers.

(d) Concepts of controllability and observability

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